Government of **Western Australia**
Curriculum Council

WACE
Western Australian Certificate of Education

Western Australian Certificate of Education Examination, 2011

Question/Answer Booklet

PHYSICS

Stage 2

Please place your student identification label in this box

Student Number: In figures

In words ______________________________________

____

Time allowed for this paper

Reading time before commencing work: ten minutes
Working time for paper: three hours

Materials required/recommended for this paper

To be provided by the supervisor
This Question/Answer Booklet
Formulae and Constants Sheet

To be provided by the candidate
Standard items: pens, pencils, eraser, correction fluid/tape, ruler, highlighters

Special items: non-programmable calculators satisfying the conditions set by the Curriculum Council for this course

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

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Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Short answers	15	15	70	64	40
Section Two: Problem-solving	7	7	90	80	50
Section Three: Comprehension	1	1	20	16	10
				Total	100

Instructions to candidates

1. The rules for the conduct of Western Australian external examinations are detailed in the *Year 12 Information Handbook 2011.* Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer Booklet.

3. Working and reasoning should be shown clearly when calculating or estimating answers.

4. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.

5. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
   - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
   - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Section One: Short answers **40% (64 Marks)**

This section has **fifteen (15)** questions. Answer **all** questions. Write your answers in the spaces provided.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

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Suggested working time: 70 minutes.

Question 1 **(3 marks)**

Medical workers are instructed to reduce their health risk from exposure to radioisotopes by following the procedures listed below. Explain how each procedure protects medical workers.

Limit time of exposure

Maximise distance from radiation sources

Wear protective clothing and use protective shielding devices

Question 2 **(3 marks)**

Answer **TRUE** or **FALSE** to each of the following.

You place a mercury thermometer and an alcohol thermometer into the same beaker of warm liquid at the same time. When the mercury and alcohol bars reach steady points:

the mercury has the same temperature as the warm liquid. ____________________

the alcohol has the same temperature as the mercury. ____________________

the mercury and alcohol have both absorbed the same amount of heat from the warm liquid.

____

Question 3 **(6 marks)**

A sudden and strong thunderstorm caused a 40.0 kg branch to break off a tree and fall from 9.00 m above the ground. The branch hit the roof of a house under the tree. Assume the branch was in free fall and the average height of the roof was 3.50 m above the ground.

(a) Calculate the speed of the branch when it hit the roof. (3 marks)

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(b) The roof was strong enough to withstand the impact and the branch settled on the roof. If the impact contact time was 0.400 seconds, calculate the magnitude of the average force exerted on the roof during the impact. (3 marks)

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Question 4 **(4 marks)**

A plastic iceblock tray containing 250 mL of water, initially at 18°C, is placed in the freezer compartment of a refrigerator. How much heat must be lost from this water when it has all become ice at 0°C?

Question 5 **(3 marks)**

A solar hot water system collects solar energy at the rate of 800 J s^{-1}. It needs 1.68×10^7 J of energy to heat the water it contains. Calculate how many hours it would take to heat this water. Ignore any heat lost to the surroundings.

Question 6 **(3 marks)**

When we go to bed on a winter's night we usually cover ourselves with a blanket, doona or quilt to keep warm. Explain briefly how this keeps us warm by referring to the three methods of heat transfer listed below.

Conduction: ____

____

____

Radiation: ____

____

____

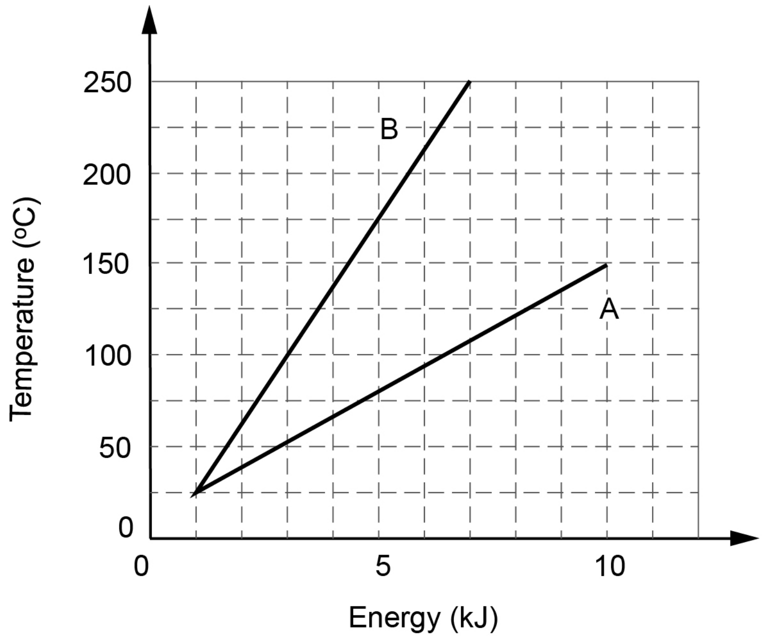
Convection: ____

____

____

Question 7 **(6 marks)**

Two objects, A and B, have the same sizes but different masses. The specific heat capacity of each is 234 J kg^{-1} K^{-1}. A and B each receive the same amount of energy from an external source. The graph shows the result.



(a) Which object has the larger mass? Using the correct mathematical formula, explain your reasoning for this choice. (3 marks)

(b) From the data in the graph, determine the mass of object A. (3 marks)

Question 8 **(5 marks)**

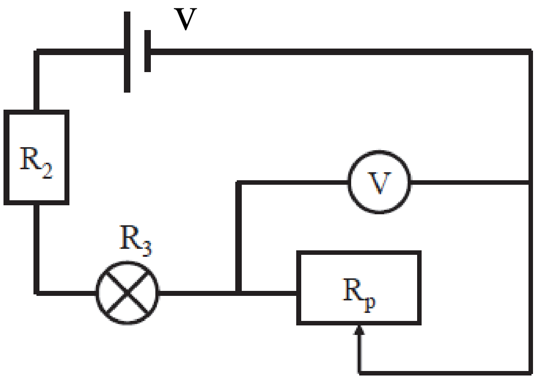
A beam of electrons in the electron gun of a cathode ray tube is accelerated from rest through a potential difference of 80 kV.

(a) How much energy in joules does each of these electrons gain in being accelerated in this field? (2 marks)

(b) If the energy gained by each electron is all kinetic, what velocity will an electron have after undergoing this acceleration? (3 marks)

Question 9 **(3 marks)**

A circuit diagram is shown below. The voltage V of the power supply is constant. R_p is a potentiometer (variable resistor). R_2 has a constant value.



When the resistance of the potentiometer (R_p) increases (the resistance of R_2 and globe R_3 remaining the same), which of the following happens (ignore the resistance of the globe)?

(i) The reading of the voltmeter increases and the globe is brighter.
(ii) The reading of the voltmeter decreases and the globe is darker.
(iii) The reading of the voltmeter increases and the globe is darker.
(iv) The reading of the voltmeter decreases and the globe is brighter.

Answer: ______________

Give reasons for your choice by completing the statements below.

The voltmeter reading ____

because ____

____

The globe is ____

because ____

____

Question 10 **(6 marks)**

A power pack for a laptop computer delivers 19.5 V with a current of 2.05 A. It is connected to the computer to recharge the battery for 2.50 hours.

(a) How much charge flows from the power pack to the battery in that time? (3 marks)

(b) How much work is done in moving this charge? (3 marks)

Question 11 **(3 marks)**

Many multi-outlet power boards are rated for a maximum current of 7.50 A. Why should we **not** connect a portable electric heater that is rated at 240 V, 2.4 kW to such a board?
Explain, showing relevant calculations.

Question 12 **(3 marks)**

Select one electrical safety device you have studied this year and explain how it works.

Safety device: ______________________________

Explanation: ______________________________

Question 13 **(7 marks)**

Calculate the average binding energy per nucleon (in joules) for carbon-14.

Use the mass of carbon-14 = 2.32478×10^{-26} kg.

Use the **Formulae and Constants sheet** for the masses of the neutron and the proton.

Question 14 **(3 marks)**

Use the mathematical relationship (formula) for momentum to explain why a slow-moving train will generally have more momentum than a high-velocity small-calibre (diameter) rifle bullet. (Calculations are not required.)

Question 15 **(6 marks)**

Sam and Henry run out of petrol not far from a service station and decide to push their car to the service station to refuel it. Sam is able to exert a force of 600 N and Henry a force of 660 N. The mass of the car is 1150 kg and there is a frictional force of 850 N. Both boys push from the rear of the car in the same direction.

(a) Using the picture below, show all of the forces, with their magnitudes, acting on the car under these conditions. (3 marks)

(b) What is the magnitude of the resultant force acting on the car under these conditions? (3 marks)

End of Section One

Section Two: Problem-solving **50% (80 Marks)**

This section contains **seven (7)** questions. Answer **all** questions. Write your answers in the spaces provided.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

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Suggested working time: 90 minutes.

Question 16 **(8 marks)**

An isotope of thorium-232 undergoes radioactive decay via a number of α and β decays. The overall decay equation can be written as:

$$^{232}_{90}\mathbf{Th} \rightarrow {}^{220}_{86}\mathbf{Rn} + \mathbf{n}\alpha + \mathbf{m}\beta$$

(a) Calculate (showing all working) the values of **n** and **m**. (4 marks)

Thorium-232 is the most commonly occurring isotope of thorium and is a solid with a half-life of around 14.05 billion years. It occurs naturally and is in plentiful supply on the Earth's surface. Radon-220 is a gas at room temperature. It is also radioactive, with a half-life of approximately 55 seconds. It decays to polonium-216, with the release of an alpha particle.

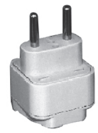
(b) Explain what is meant by 'the half-life of radon-220 is 55 seconds'. (2 marks)

(c) Even though radon gas has a very short half-life and is not abundant, it is more dangerous to humans than thorium. Explain why. (2 marks)

Question 17 **(14 marks)**

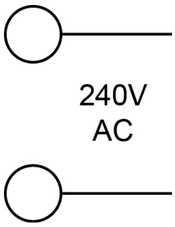
Adam returns from Singapore with a lamp only to find that the plug on the lamp does not fit the power outlets in Australia. The lamp consists of a plug, a switch and a globe. The tag on the lamp says it is designed for a 110 V power supply. The Australian power supply voltage is 240 V.

Lamp



Lamp plug

(a) Using appropriate circuit symbols complete the circuit diagram for this lamp when connected to the 240V AC power supply. (3 marks)



The globe in the lamp is marked ‘110 V, 0.60 A’.

(b) What is the power rating of this globe when it is used in Singapore? (2 marks)

(c) What is the theoretical resistance of this globe under these conditions? (2 marks)

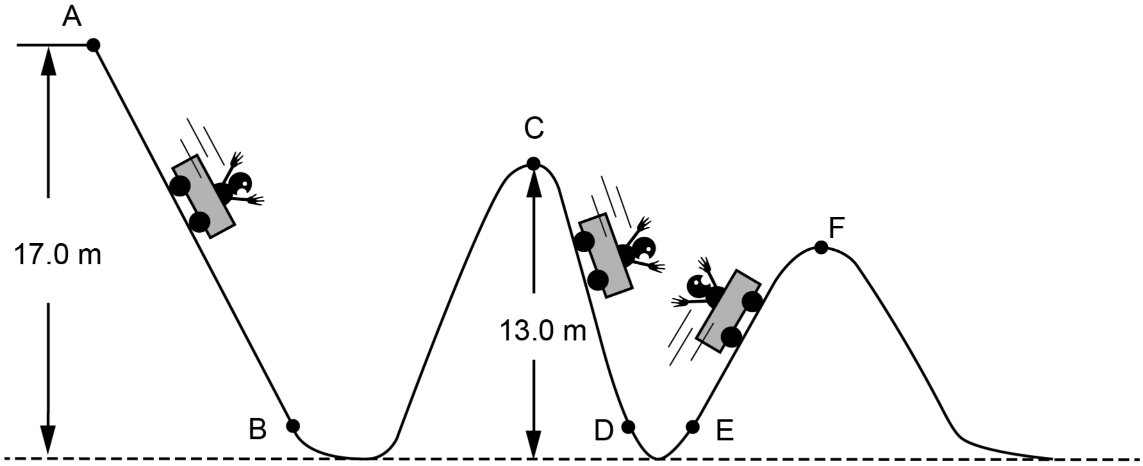
(d) Adam buys a plug suitable for Australian power outlets, cuts the Singaporean plug from the wires and connects the Australian plug. Assuming that he has wired it correctly, is the lamp now safe to use in Australia? Explain. (2 marks)

(e) The original 110 V globe is changed to an Australian-made globe with a power rating of 75 W. What is the theoretical resistance of the new globe when it is connected to the Perth electricity supply? (3 marks)

(f) In (c) you were asked to calculate the theoretical resistance. This is because an operating globe is a non-ohmic conductor. Explain what is meant by the term ‘non-ohmic conductor’. (2 marks)

Question 18 **(14 marks)**

A roller-coaster is 17.0 m high at its highest point, the release point A. The diagram below is a simple representation of the first part of the ride.



(a) The car is released from a stationary position at A and has no independent locomotion. Indicate the direction of the car's acceleration in each of the following regions by circling the correct answer. (3 marks)

(i) A to B up slope down slope

(ii) C to D up slope down slope

(iii) E to F up slope down slope

(b) If the length of track between A and B is 21.0 m and the car is released from rest at A and reaches a velocity of 18.5 m s^{-1} at B, what is the magnitude of the acceleration it experiences between A and B? (4 marks)

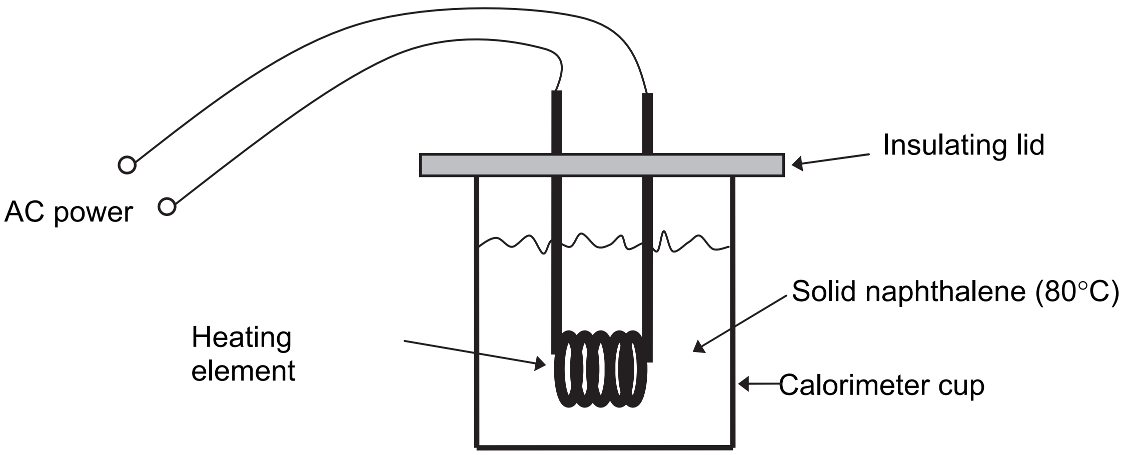
(c) If the car **just** makes it to point C, what proportion (fraction) of the energy has been lost as heat? (3 marks)

(d) In relation to this energy lost as heat, how is it generated and where does it go? (2 marks)

(e) Will the velocity of the car be greater at point B or point D? Give a reason for your answer. (2 marks)

Question 19 **(17 marks)**

Hayley investigated the accuracy of the stated power rating on a heating element that was marked '100 W'. She placed solid naphthalene at its melting point of 80.0°C in a calorimeter cup at 80.0°C together with the heating element, and timed how long it took to melt varying amounts of naphthalene.



She collected the following results:

Mass of naphthalene (kg)	Time to melt (s)
0.150	288
0.250	480
0.350	670
0.450	865
0.550	1055

L_f (naphthalene) = 1.48×10^5 J kg^{-1}

She then used the following relationship to decide how to plot the data:

$$P = \frac{Q}{t} = \frac{mL_f}{t}$$

She used the gradient of the line from her plot to calculate the actual power output of the heating element.

(a) Draw a clearly-labelled graph on the graph paper provided on page 19. Plot time on the x-axis and mass on the y-axis. (5 marks)

If you wish to make a second attempt at this item, the graph is repeated on page 28 of this booklet. Indicate clearly on this page if you have used the second graph and cancel the working on the graph on this page.

(b) From the graph, obtain the gradient of the line you have plotted. Show all workings and include the units in your answer. (5 marks)

(c) Use the gradient of the graph and the equation to calculate the power output of the heating element. (3 marks)

(d) From the graph, determine the time taken to melt 0.300 kg of naphthalene. (2 marks)

(e) Would you expect the calculated power rating to be higher or lower than the true power rating (not the rating printed on the heating element)? Explain your answer. (2 marks)

Question 20 **(10 marks)**

A flea's jump is one of the most impressive examples of acceleration in the animal kingdom. By pushing its legs against the ground, the flea can attain an initial upward velocity of 1.00 m s^{-1} in 10.0 milliseconds.

Calculate:

(a) the flea's average acceleration over this time (that is, while leaving the ground). (3 marks)

(b) the force acting on the flea during this time if it has a mass of 2.00 mg. (4 marks)

(c) the maximum height that the flea can reach if its initial velocity is vertically upward. Hint: once the flea leaves the ground, it is affected only by gravity. (3 marks)

Question 21 **(10 marks)**

Technetium-99m (Tc-99m) has a half-life of 6 hours and emits only gamma rays. It is used as a radioactive tracer to diagnose illness, which means it is swallowed and collects in the part of the body that needs to be seen. A detector is then used to form an image of that part of the body.

(a) Select a characteristic of Tc-99m that makes it a good choice for use as a radioactive tracer and explain the benefit of this characteristic. (2 marks)

Characteristic: ____________________

Explanation: ____________________

____

(b) The initial activity of a sample used as a radioactive tracer is 320 Bq. While no activity is safe, it is considered that an activity of 5.00 Bq gives a significantly-reduced risk. How long after swallowing can a patient expect to wait to achieve a significantly-reduced risk? (3 marks)

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(c) The energy released to a particular body organ of mass 3.00 kg that Tc-99m collects in is 3.10×10^{-2} J in one minute. What is the absorbed dose in this time? (3 marks)

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(d) If a different radioisotope that emitted only alpha radiation was used by mistake to produce the same absorbed dose, it would have a much higher dose equivalent. Give an explanation for this difference. (2 marks)

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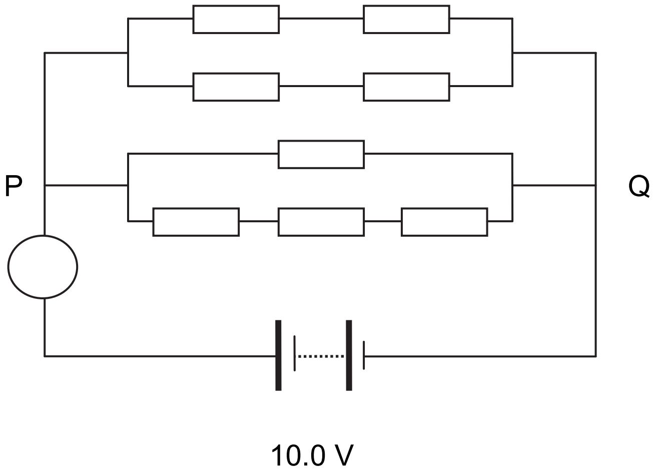
Question 22 **(7 marks)**

Eight equal value resistances are connected between P and Q. The resistance of each of these resistors is 10.0 Ω.



(a) Calculate the total resistance of the circuit, excluding the meter and the power source. (5 marks)

(b) The circle represents a meter. What is the reading on this meter (include the unit)? (2 marks)

End of Section Two

Section Three: Comprehension **10% (16 Marks)**

This section contains **one (1)** question. You must answer this question. Write your answer in the spaces provided.

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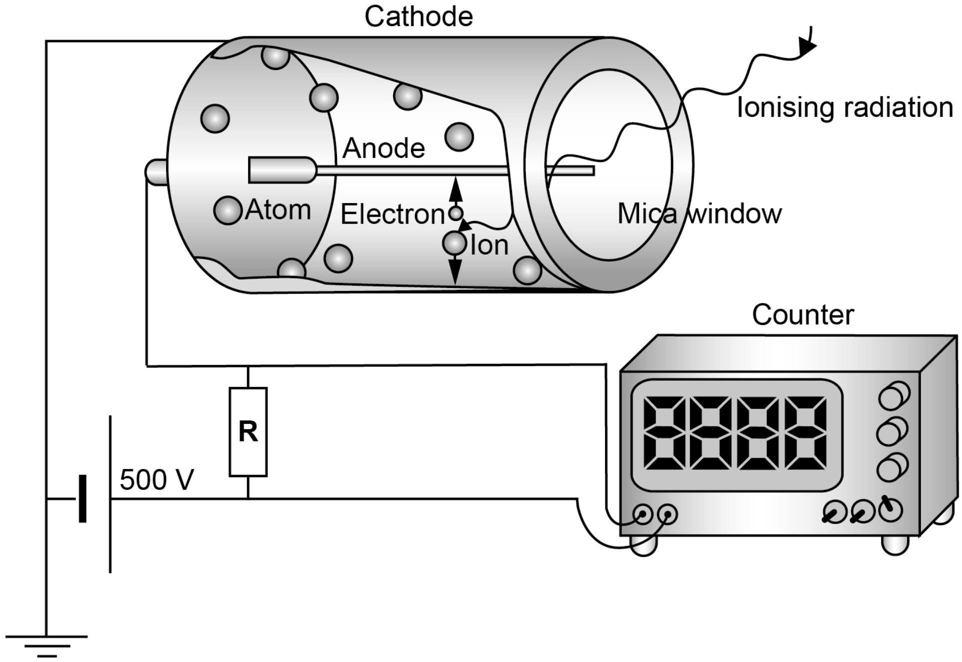
Suggested working time: 20 minutes.

Question 23

A Geiger–Müller tube (or GM tube) is the sensing element of a Geiger counter instrument that can detect a single particle of ionising radiation.



A GM tube consists of a tube filled with a low-pressure inert gas such as helium, neon or argon. The tube contains electrodes, between which there is a potential difference of several hundred volts, but no current flowing. The walls of the tube are either entirely metal or have their inside surface coated with a conductor to form the cathode while the anode is a wire passing up the centre of the tube.

When ionising radiation passes through the tube, some of the gas molecules are ionised, creating positively charged ions and electrons. The strong electric field created by the tube's electrodes accelerates the ions towards the cathode and the electrons towards the anode. The ion pairs gain sufficient energy to ionise further gas molecules through collisions on the way, creating an avalanche of charged particles. This results in a short, intense pulse of electrons that passes (or cascades) from the negative electrode to the positive electrode and is measured or counted.

The number of pulses per second measures the intensity of the radiation present. Some Geiger counters display an exposure rate, but this does not relate easily to a dose rate as the instrument does not discriminate between radiations of different energies.

See next page

The usual form of tube is an end-window tube. This type is so-named because the tube has a window at one end through which ionising radiation can easily penetrate. The other end normally has the electrical connectors. There are two types of end-window tubes: the glass-window type and the mica (which can be split into very thin sheets) window type. The glass-window type will not detect alpha radiation since it is unable to penetrate the glass, but is usually cheaper and will usually detect beta radiation and gamma rays. The mica window type will detect alpha radiation but is more fragile.

(a) Explain how the ionising radiation causes the inert gas inside the GM tube to become charged. (2 marks)

(b) On the circuit diagram on page 24, draw an arrow to indicate the direction of conventional current flow. (1 mark)

(c) With reference to the properties of alpha particles, infer why a mica window will allow the detection of alpha particles while a glass window will not.

(2 marks)

(d) During one experiment, the detector indicates 648 counts over a period of 3 minutes when located 10 cm from a source.

(i) Before the activity of the source can be calculated accurately, the background count is needed. What is meant by the 'background count' and how is it produced? (2 marks)

(ii) If the background count is 14 counts per minute, calculate the source's activity. (3 marks)

(iii) As the GM tube moves closer to the source, what would you expect the counter to indicate? (2 marks)

Circle the correct answer and give reasons for your choice:

more activity less activity same activity

(e) During another experiment, 5.6×10^{12} electrons were moved through the circuit over 0.02 seconds when detecting a radioactive source. Calculate the current in the circuit for this event. (2 marks)

(f) Give two reasons why this device can determine a decay rate, but not a dose rate. (2 marks)

One:

Two:

End of questions

Additional working space

Additional working space

Additional working space

ACKNOWLEDGEMENTS

Section One:

Question 15 Photograph of car. Used with kind permission of the examining panel.

Section Two:

Question 17 Photographs of lamp and lamp plug. Used with kind permission of the examining panel.

Section Three:

Question 23 Adapted from: Knott, T. (2005, October 18). *Geiger–Müller tube* [Drawing]. Retrieved August 25, 2010, from: http://en.wikipedia.org/wiki/Geiger-M%C3%BCller_tube. This file is licensed under the Creative Commons Attribution-Share Alike 3.0 Unported license.

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